Group Regression Project Final

Team 2:

- Akash Taploo
- Raj Jasani
- Sameer Bhattarai
- Sneha Mondal

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Professor Jing Tang
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1. Executive Summary

This report presents the results of a study conducted for the American Cancer Society to investigate the various factors affecting cancer incidence rates in the United States. The study utilized data from the American Cancer Society and included key factors such as cancer incidence rate, five-year trends, clinical trials, and states. Through regression analysis, we found that recent trends and study counts significantly impact the incidence rate and identified three states (Kentucky, Delaware, and New York) as most vulnerable to cancer. Additionally, our analysis revealed that median income is significantly linked to cancer incidence rates, with higher median income associated with lower incidence rates. Our findings provide valuable insights for the American Cancer Society to further examine and address the factors contributing to cancer incidence rates in the US.

Finally, we developed regression models to predict the impact of independent factors on the incidence and death rates, providing a comprehensive understanding of the potential behaviour of dependent variables. This study demonstrates the importance of continued research and intervention to combat the significant impact of cancer on individuals and communities across the United States.

2. Introduction

Cancer is a devastating illness that affects millions of individuals worldwide, and the United States is no exception. While advancements in prevention, diagnosis, and treatment have led to a decrease in cancer incidence rates, it remains the second leading cause of death in the US. Various risk factors have been identified as contributors to cancer development, many of which are modifiable. By understanding the factors that contribute to cancer incidence rates, targeted interventions can be developed to reduce the burden of cancer on individuals and communities.

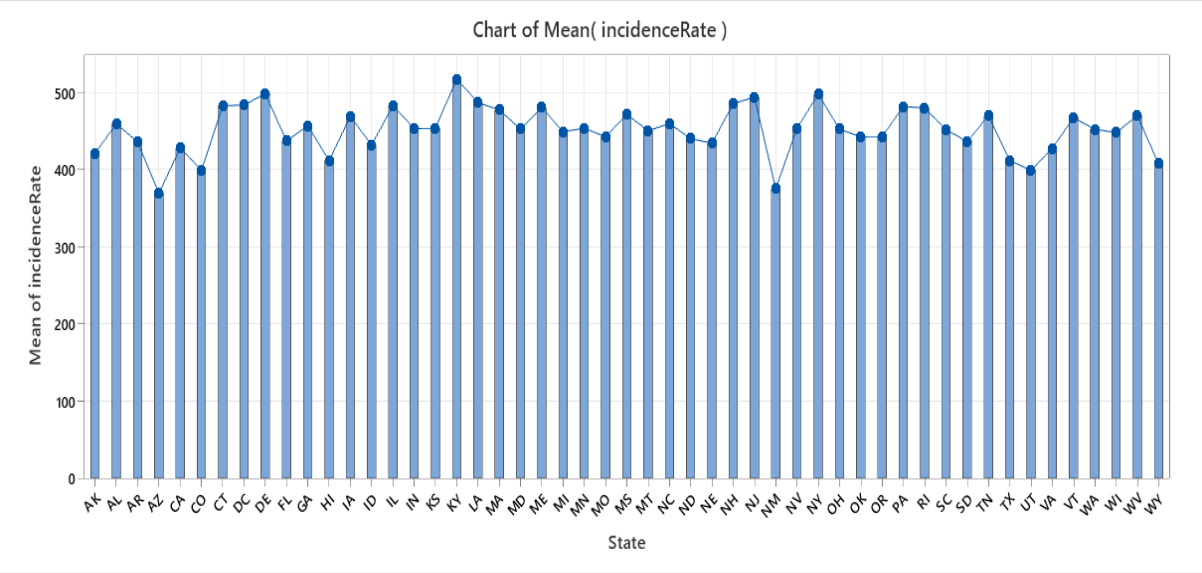
The purpose of this study, conducted for the American Cancer Society, is to explore the factors related to cancer deaths in the United States. Using data from all zip codes in the country, we aimed to create a model of cancer incidence rates and identify regions and associated partners that require interventions. Through our analysis, we hope to provide insights that will inform effective strategies for cancer prevention and control in the US.

Our team compiled data from a variety of reliable sources to analyse the incidence of cancer in various counties. We utilized publicly accessible state and national data to gather information on population size, poverty rates, and median income, which we then aggregated by county. In addition, we gathered information from the National Cancer Institute's Surveillance, Epidemiology, and End Results program, which provided data on cancer incidence rates, average annual cancer counts, recent and five-year trends in incidence rates, cancer death rates, average deaths per year, and recent trends in deaths. We also utilized demographic information provided by the United States Census Bureau to further understand the relationship between cancer incidence rates and other socio-economic factors. By combining and analysing these various sources of data, we aim to better understand the factors contributing to cancer incidence rates at the county level and identify areas for targeted interventions and prevention strategies.

The regions were established at both the county and state levels. We also compared the incidence rate to the median income by categorizing it into four groups: Very Low, Low, High, and Very High. The counties with the greatest and lowest median income and incidence rate in each of the four categories were then determined. Finally, in order to estimate the incidence rate, we found the parameters having the strongest association to it and used them to create a statistical regression model.

3. Regions Most Prone to Cancer

To give our customer, the American Cancer Society, with significant insights about the cancer incidence rate in the United States, we first identified cancer-prone regions. We opted to discover measures of central placement such as Mean and Median (Appendix, Table 1) of incidence rate by state in order to obtain the information of interest. In our example, central location measures are more relevant than absolute value of counts as a measure of items in a subset since they reflect the average of elements in a subset. Using the statistical program Minitab, we developed a chart that shows the mean incidence rate by state, from greatest to lowest.



From the above graphs, we conclude, Kentucky (517.3) has the highest average incident rate which is followed by Delaware (498.2), and New York (497.5) whereas, Arizona (369.3) has the lowest incident rate followed by New Mexico (370.9), and Utah (396.7)

As pet the United States Census Bureau, the states are classified into four regions: Northeast, Midwest, South, and West.
According to the United States Census Bureau, the states are classified into four major regions: Northeast, Midwest, South, and West. The two states with the greatest average incidence rate (KY and DE) are in the South, whereas the three states with the lowest rate (AZ, NM and UT) are in the West. As a result, we may infer that the South is the most cancer-prone area in the United States, while the West is the least. With these findings, we may recommend additional investigation of these two locations to determine the key factors that may influence their distinction in terms of cancer incidence rate.

4. RecentTrend & StudyCounts affecting incidenceRate

We ran a hypothesis test to see if factors such as ***RecentTrend*** and ***StudyCounts*** substantially impact the subject of the study to construct a credible model of the cancer incidence rate in the United States. The results of these tests offer preliminary information about the variables' effect and help us to determine if the variables should be investigated further.
The hypothesis was created as the initial stage in the testing procedure. In our example, we constructed the following two hypotheses in line with well acknowledged rules:

$$\mathbf{H_0: u_1 - u_2 = 0}$$
$$\mathbf{H_a: u_1 - u_2 \neq 0}$$

- The null hypothesis states that there is no significant difference in the population mean of Incidence Rate when factor is applied.
- The alternative hypothesis states that there a difference in the population mean of Incidence Rate when factor is applied.

(A) Recent Trend *(stable vs falling)*

In accordance with the American Cancer Society's request, we purified the submitted "cancer death" data, removing unknown and "increased" trend categories and replacing them with "***falling***" and "***stable***" ones and developed out hypothesis as follows:

$$\mathbf{H_o: d = u_1 - u_2 = 0}$$
$$\mathbf{H_a: d = u_1 - u_2 \neq 0}$$

u_1 is population mean of incidence rate when recent trend is falling, and

u_2 is population mean of incidence rate when recent trend is stable.

- The null hypothesis states that there is no difference between the population mean of Incidence Rate when Recent Trend is falling and when Recent Trend is stable.
- The alternative hypothesis states that there is a difference between the population mean of Incidence Rate when Recent Trend is falling and when Recent Trend is stable.

In the Two Sample t-test, we set the confidence interval is set 95% to increase the reliability of our results. This number is critical in determining the significance level of the variable ***Recent Trend*** on cancer ***incidenceRate*** when compared to the ***P-value***. We obtained *P-value* equals to 0.000 after doing the hypothesis test **(Figures 1.1 and 1.2)**. To be considered significant, the ***P-value*** of the test should be smaller than the ***alpha value*** of 0.05, according to statistical technique. Thus, having observed that our ***P-value*** is smaller than the ***alpha value***, we rejected the Null Hypothesis and inferred that the "Recent Trend" is an essential element impacting the population mean "Incidence Rate".

Method

μ_1: population mean of incidenceRate when recentTrend = falling
μ_2: population mean of incidenceRate when recentTrend = stable
Difference: $\mu_1 - \mu_2$

Equal variances are not assumed for this analysis.

Descriptive Statistics: incidenceRate

recentTrend	N	Mean	StDev	SE Mean
falling	6956	448.6	42.1	0.51
stable	23381	455.4	47.9	0.31

(Figure 1.1)

Estimation for Difference

Difference	95% CI for Difference
-6.797	(-7.962, -5.631)

Test

Null hypothesis	$H_0: \mu_1 - \mu_2 = 0$
Alternative hypothesis	$H_1: \mu_1 - \mu_2 \neq 0$

T-Value	DF	P-Value
-11.43	12767	0.000

(Figure 1.2)

To support our theory, we displayed our data in form of a histogram, separating it into two categories of "***falling***" and "***stable***". The population mean of ***incidenceRate*** for the falling group is 448.83 and for the stable group is 455.66. **(Figure 1.3)**

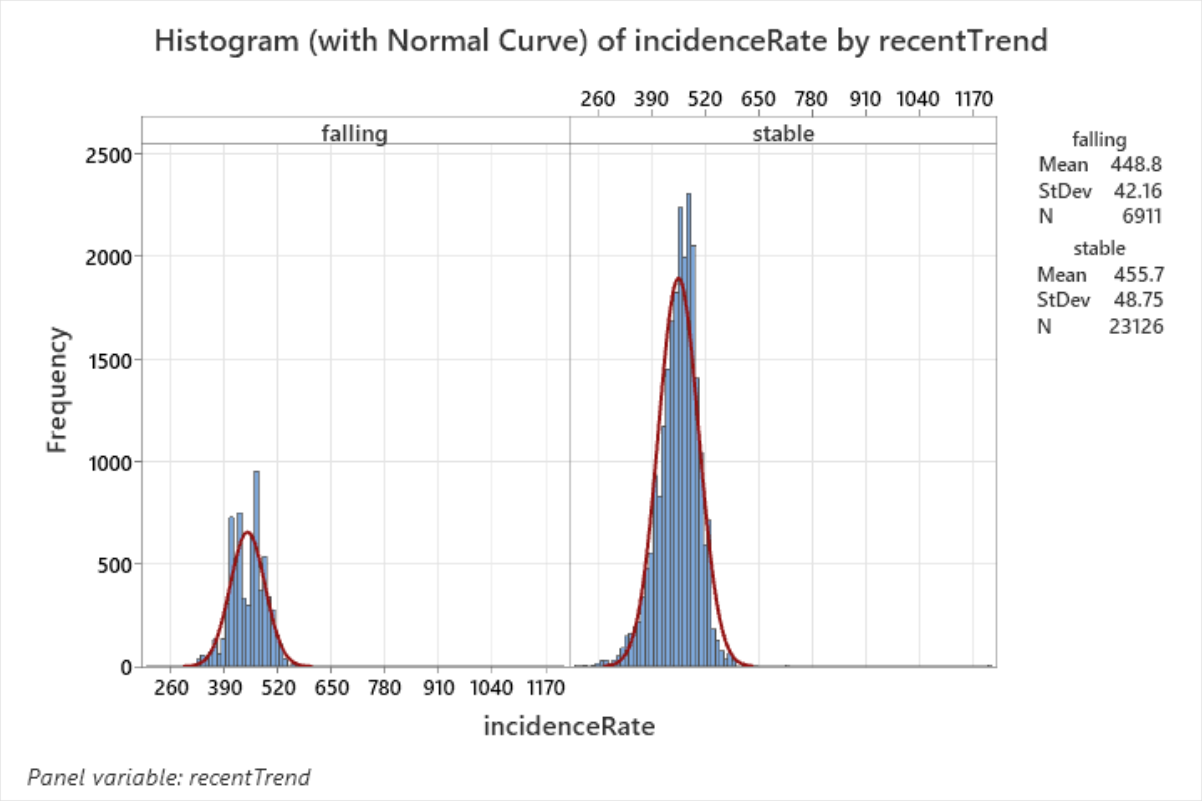


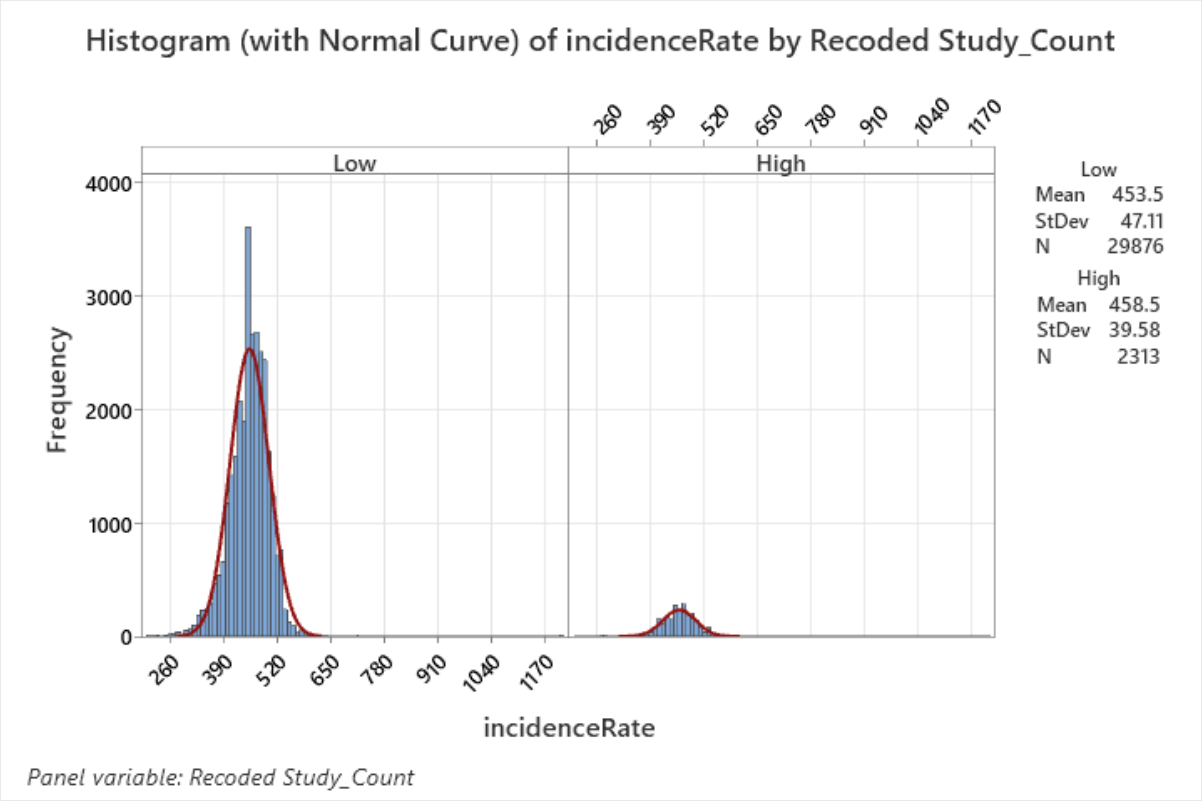
Figure (1.3)

(B) Study Counts *(low vs high)*

To continue examining possible factors that may influence the ***incidenceRate***, we used the ***StudyCounts*** data provided to us to perform a hypothesis test. We updated (recoded) our dataset by categorizing data from ***StudyCounts*** into two categories of *"**Low**"* and *"**High**"* where the range0 to 2.5 fell within the low category and 2.5 to 1535.1 fell within the high category.

$$H_0: d = u_1 - u_2 = 0$$
$$H_a: d = u_1 - u_2 \neq 0$$

- The null hypothesis states there is no difference between the population mean of *Incident Rate* when *Study Counts* is low and when it is high.
- The alternative hypothesis states there is a difference between the population mean of *Incident Rate* when *Study Counts* is low and when it is high.



The above histogram is a graphical illustration of the ***incidenceRate*** using ***Recoded StudyCounts***. We can conclude that the group with a greater number of study counts has a lower frequency of incidence rate compared to the group with lesser study counts. Furthermore, we can reject our null hypothesis since the population mean for Low Study Count group (453.5) differs from High Study Counts group (458.5)

To draw further inferences, we looked at the results of ***Two Sample T-test***. The ***P-value*** for ***StudyCounts*** was 0.00, which is smaller than the ***alpha value*** *of 0.05*, which allows us to reject the null hypothesis and conclude that ***StudyCounts*** influences ***incidenceRate***.

5. ANOVA test of Median Incomes

To evaluate how the variable of median incomes relate to the cancer data, we analysed it through a four-level indicator which is provided in the table below:

Lower End	Upper End	Recoded MedIncome	Number of Rows
22640	41561.1	Very Low	7587
41561.1	48264.1	Low	7609
48264.1	55915.1	High	7569
55915.1	125635.1	Very High	7576

To ensure the reliability of our ANOVA test, we performed a test for equal variances by constructing the following hypothesis.

H_0: All variants are the same.
H_a: Not all variants are same.

The test **(Figure 2)** yielded ***P-value*** of 0.00, which is less than the ***alpha value*** of 0.05. This finding demonstrated that the variations between the four levels of Median Income are not identical and thus we shouldn't assume equal variances for ANOVA test.

Test for Equal Variances: incidenceRate versus Recoded medIncome

Method

Null hypothesis	All variances are equal
Alternative hypothesis	At least one variance is different
Significance level	α = 0.05

95% Bonferroni Confidence Intervals for Standard Deviations

Recoded medIncome	N	StDev	CI
Very Low	7587	56.8292	(53.8281, 60.0173)
Low	7609	44.5944	(43.0655, 46.1927)
High	7569	42.4668	(41.5031, 43.4672)
Very High	7576	44.2793	(43.1349, 45.4691)

Individual confidence level = 98.75%

Tests

Method	Test Statistic	P-Value
Multiple comparisons	—	0.000
Levene	131.84	0.000

(Figure 2)

Further, we conducted the One-Way ANOVA test (***Incidence Rate vs Recoded Median Income***), which gave us the ***P-value*** of 0.00 which is smaller than the *alpha value* allowing us to conclude that the means for all the levels differ from each other. (Refer to **Figures 3.1 and 3.2** below)

One-way ANOVA: incidenceRate versus Recoded medIncome

Method

Null hypothesis	All means are equal
Alternative hypothesis	Not all means are equal
Significance level	α = 0.05

Equal variances were not assumed for the analysis.

Factor Information

Factor	Levels	Values
Recoded medIncome	4	Very Low, Low, High, Very High

Welch's Test

Source	DF Num	DF Den	F-Value	P-Value
Recoded medIncome	3	16779.9	17.13	0.000

(Figure 3.1)

Model Summary

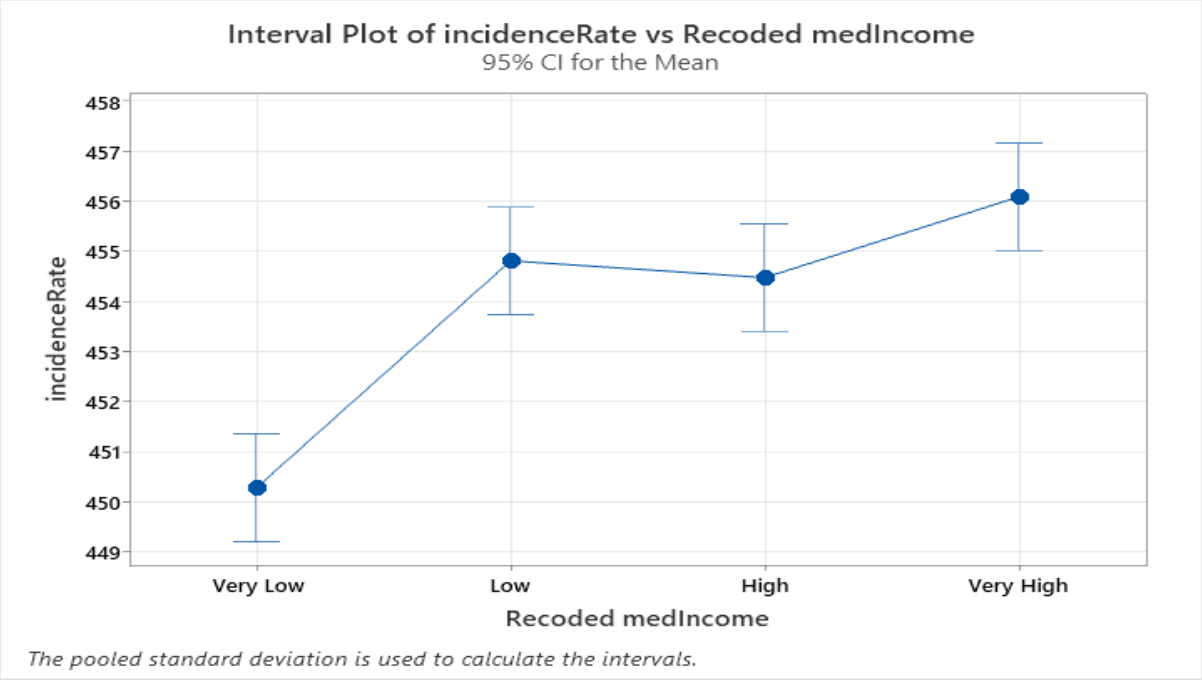
R-sq	R-sq(adj)	R-sq(pred)
0.21%	0.20%	0.19%

Means

Recoded medIncome	N	Mean	StDev	95% CI
Very Low	7587	450.274	56.829	(448.995, 451.553)
Low	7609	454.802	44.594	(453.800, 455.804)
High	7569	454.468	42.467	(453.511, 455.424)
Very High	7576	456.080	44.279	(455.083, 457.078)

(Figure 3.2)

Graphical representation of the means of ***Recoded MedIncome*** at four levels:



In addition, when the condition of variance homogeneity was broken, we carried out the ***Games-Howell test*** in order to compare all the possible combinations of group differences and the results are pasted below:

Games-Howell Pairwise Comparisons

Grouping Information Using the Games-Howell Method and 95% Confidence

Recoded medIncome	N	Mean	Grouping	
Very High	7576	456.080	A	
Low	7609	454.802	A	
High	7569	454.468	A	
Very Low	7587	450.274		B

Means that do not share a letter are significantly different.

Games-Howell Simultaneous Tests for Differences of Means

Difference of Levels	Difference of Means	SE of Difference	95% CI	T-Value	Adjusted P-Value
Low - Very Low	4.528	0.829	(2.400, 6.655)	5.46	0.000
High - Very Low	4.193	0.815	(2.102, 6.285)	5.15	0.000
Very High - Very Low	5.806	0.827	(3.683, 7.930)	7.02	0.000
High - Low	-0.335	0.707	(-2.149, 1.480)	-0.47	0.965
Very High - Low	1.278	0.721	(-0.573, 3.130)	1.77	0.287
Very High - High	1.613	0.705	(-0.197, 3.423)	2.29	0.101

According to the grouping information, Group A has Very High, Low, and High, but Group B contains just Very Low. There is nothing in common between the two groups. The amounts of factors within a group are not significantly different from one another. Because the letter is not shared by all four groups, they are all notably distinct. Moreover, combinations with an ***Adjusted P-value*** less than 0.05 suggest that the difference between all of these means is significant. The first three options in our scenario fall under this group.

6. Multicollinearity Problem

Multicollinearity is statistical problem that arises when there is a strong intercorrelation between two or more independent variables. This means that when independent variables have a significant impact on each other, it becomes difficult to identify the true relationship between the independent variables and the dependent variable. As a result, the conclusions drawn from the analysis become ambiguous, and the standard error increases, making statistical models less reliable.

To address this problem, we employed various statistical techniques and tools. One of the tools we used was the ***Pearson Coefficient Matrix***, which provided us with a table of intercorrelations between continuous variables (excluding categorical ones). We set a threshold of 0.7 for the sample correlation coefficient between two independent variables to identify the pairs of variables that were strongly correlated. As we can see, several pairs of variables exceed the 0.7 correlation threshold (highlighted in **Figure 4** below)

Correlations

	incidenceRate	studyCount	PovertyEst	povertyPercent	medIncome	popEst2015	avgAnnCount	fiveYearTrend	deathRate
studyCount	0.019								
PovertyEst	-0.102	0.086							
povertyPercent	-0.069	-0.003	0.067						
medIncome	0.046	0.050	0.114	-0.770					
popEst2015	-0.088	0.090	0.989	-0.004	0.208				
avgAnnCount	-0.040	0.095	0.970	-0.028	0.237	0.989			
fiveYearTrend	0.175	-0.006	-0.055	0.043	-0.064	-0.058	-0.054		
deathRate	0.444	-0.029	-0.191	0.441	-0.496	-0.228	-0.224	0.099	
avgDeathsPerYear	-0.045	0.094	0.973	-0.014	0.213	0.987	0.999	-0.054	-0.209

(Figure 4)

Based on our analysis, we found that the ***PovertyEst*** variable had a better correlation coefficient with the dependent variable, Incidence Rate (-0.102), than the other variables. Therefore, we decided to keep ***PovertyEst*** and remove the other variables from the model. It helps to remove multicollinearity issues from the model.

To confirm the presence of multicollinearity, we also used the Variance Inflation Factor (VIF), which measures the degree of correlation between independent variables by calculating the degree to which the variance of the calculated regression coefficients is exaggerated. Based on our analysis (as shown in **Figure 5**), we found that several variables had VIF scores greater than 5, indicating significant correlation between the variables. Based on the table, we may infer that several of the variables with VIF scores larger than 5 (highlighted in **Figure 5** below) are strongly linked and may have a detrimental impact on the dependability of our model. As a result, we acquired further evidence to remove all of them except PovertyEst (since it has a greater correlation coefficient with the Incidence Rate).

Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	261.32	2.92	89.36	0.000	
studyCount	0.0439	0.0104	4.21	0.000	1.01
PovertyEst	0.000238	0.000010	25.03	0.000	94.09
povertyPercent	-1.9798	0.0597	-33.18	0.000	2.81
medIncome	0.000276	0.000030	9.25	0.000	3.99
popEst2015	-0.000121	0.000002	-49.67	0.000	195.66
avgAnnCount	0.07289	0.00124	58.64	0.000	788.76
fiveYearTrend	1.8123	0.0668	27.14	0.000	1.02
deathRate	1.1998	0.0101	118.45	0.000	1.45
avgDeathsPerYear	-0.14247	0.00307	-46.43	0.000	645.01

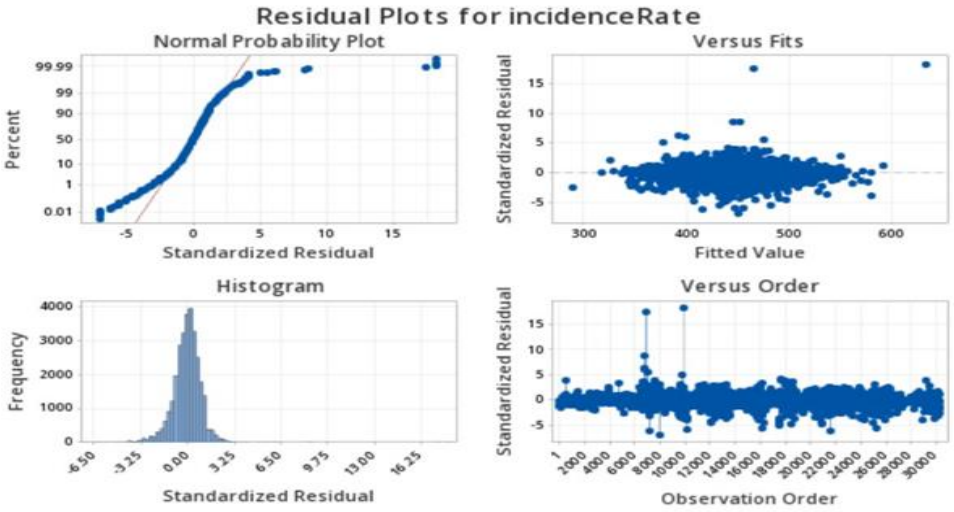
(Figure 5)

7. Regression Models

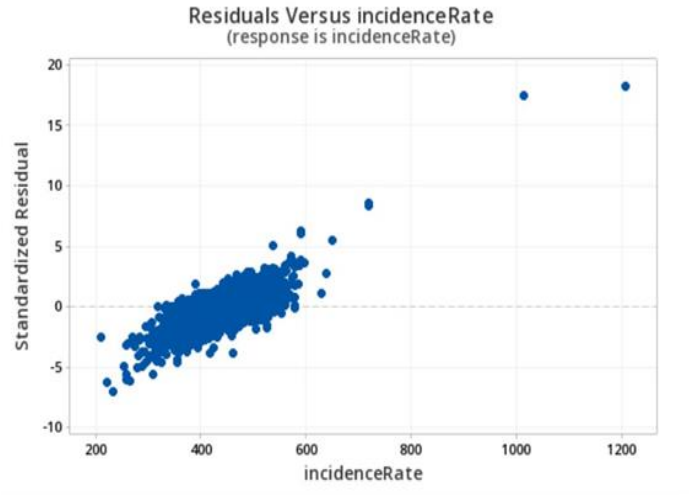
To build an appropriate model, we used a step-by-step procedure that adhered to well acknowledged statistical norms.

We started our study with just important variables to reduce the chance of misleading outcomes and developed a coefficient table to identify inter-correlated independent variables with higher VIF scores (Appendix, Table 2).

In addition, by creating "Residual Plots of Incidence Rate" **(Figures 6.1 and 6.2)**, we were able to detect outliers that might undermine the credibility of our model.



(Figure 6.1)



(Figure 6.2)

We then eliminated highly correlated variables and variables with multicollinearity difficulties (citing high VIF score), removed certain outliers, and introduced categorical variables such as ***State***, ***Recoded StudyCounts***, and ***Latest Trend***. After these adjustments, we obtained preliminary data for our model **(Figures 7.1, 7.2 and 7.3)**. Later, we were able to develop our Regression Model Equation for Cancer Incidence Rate **(Figure 7.4)** and make several key findings. It should be noted that the model meets the regression model's assumptions, such as continuous variables, normal distribution, no threat of multicollinearity, and no threat of influential instances.

Method

Categorical predictor coding (1, 0)

Continuous Predictor Standardization

Subtract the mean, then divide by the standard deviation

Predictor	Mean	StDev
medIncome	50673.2	13872
PovertyEst	70554.9	211694
fiveYearTrend	-1.8	3
deathRate	175.5	25
studyCount	2.5	20

(Figure 7.1)

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
31.8807	53.45%	53.37%	53.27%

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	53	35341164	666814	656.07	0.000
medIncome	1	2007545	2007545	1975.20	0.000
PovertyEst	1	12865	12865	12.66	0.000
fiveYearTrend	1	304545	304545	299.64	0.000
deathRate	1	8018731	8018731	7889.52	0.000
studyCount	1	31331	31331	30.83	0.000
State	47	15065681	320546	315.38	0.000
recentTrend	1	77	77	0.08	0.783
Error	30281	30776915	1016		
Lack-of-Fit	5425	30776915	5673	*	*
Pure Error	24856	0	0		
Total	30334	66118079			

(Figure 7.2)

We rejected the null hypothesis since the ***P-value*** is less than 0.05, indicating that the regression model is significant. Because all continuous variables have a *P-value* less than 0.05, they are all significant independent variables. Moreover, the co-efficient of determination i.e., ***R-Sq (Adj)*** is 53.37% which indicates that 53.37% of variances in cancer incidences are explained in this model.

Correlations

	incidenceRate	studyCount	PovertyEst	medIncome
studyCount	0.020			
PovertyEst	-0.103	0.086		
medIncome	0.048	0.050	0.114	
fiveYearTrend	0.169	-0.006	-0.055	-0.064

(Figure 7.3)

Incidence Rate:

Regression Equation in Uncoded Units

incidenceRate = 185.66 + 0.000830 medIncome + 0.000004 PovertyEst + 1.1164 fiveYearTrend
+ 0.9727 deathRate + 0.05109 studyCount + 0.0 State_AK + 55.49 State_AL
+ 26.99 State_AR + 1.87 State_AZ + 41.87 State_CA + 32.07 State_CO
+ 86.93 State_CT + 60.98 State_DC + 92.76 State_DE + 51.15 State_FL
+ 62.27 State_GA + 37.07 State_HI + 75.32 State_IA + 56.94 State_ID
+ 77.90 State_IL + 46.54 State_IN + 86.63 State_KY + 75.06 State_LA
+ 77.36 State_MA + 42.53 State_MD + 80.69 State_ME + 54.81 State_MI
+ 40.80 State_MO + 58.72 State_MS + 77.07 State_MT + 68.16 State_NC
+ 57.48 State_ND + 49.98 State_NE + 86.45 State_NH + 84.56 State_NJ
+ 7.27 State_NM + 100.95 State_NY + 48.79 State_OH + 34.27 State_OK
+ 54.67 State_OR + 83.80 State_PA + 83.17 State_RI + 52.49 State_SC
+ 55.94 State_SD + 57.26 State_TN + 24.82 State_TX + 36.79 State_UT
+ 26.08 State_VA + 72.48 State_VT + 59.62 State_WA + 54.80 State_WI
+ 54.43 State_WV + 24.05 State_WY + 0.0 recentTrend_falling
- 0.148 recentTrend_stable

(Figure 7.4)

The above multiple regression equation predicts the incident rate of cancer based on several independent variables:

1. ***medIncome***: median income
2. ***PovertyEst*:** estimated percentage of population living below poverty line.
3. ***fiveYearTrend***: five-year trend in the incidence rate.
4. ***deathRate***: death rate from cancer.
5. ***studyCount***: number of studies conducted in the area.
6. ***State***: indicates the state in which the area of study is located.
7. ***recentTrend_falling***: a variable indicating whether the recent trend is falling.
8. ***recentTrend_stable***: a variable indicating whether the recent trend is stable.

Based on these findings, we conclude that ***MedIncome*** has a positive relationship with the incidence rate as for every $1, additional increase in the median household income the incidence rate is expected to by 0.000830%. However, there is no significant relationship between the ***PovertyEst*** and the incidence rate. The slope coefficient of 0.000004 suggests that as the poverty estimate increase by 1%, the Cancer Incident rate is predicted to increase by 0.000004% holding other variables constant. The incidence ***fiveYearTrend*** has a significant positive relationship with the incidence rate. As the five-year trend increases by 1 year, the Incident rate is expected to increase by 1.1164%, holding other variables constant. There is a positive relationship between the ***deathRate*** and the incidence rate. The slope coefficient of 0.9727 suggests that as the death rate increase by 1%, the Cancer Incident rate is predicted to increase by 0.97% holding other variables constant. There is a positive relationship between the ***studyCount*** and the incidence rate. The slope coefficient of poverty percent (0.05109) indicates that for each 1 increase in clinical trials, the Incident rate is predicted to increase by 0.05109%.

Death Rate:

We reject the null hypothesis since the *P-value* is less than 0.05, indicating that the regression model is significant. Because all continuous variables have a *P-value* less than 0.05, they are all significant to the independent variable **(Figures 8.1, 8.2 and 8.3)**. The ***R-Sq (Adj)*** of 63.39% explains that 63.39% of variation in death rates for cancer is explained by this model.

Method

Categorical predictor coding (1, 0)

Continuous Predictor Standardization

Subtract the mean, then divide by the standard deviation

Predictor	Mean	StDev
studyCount	2	20
PovertyEst	70562	211707
povertyPercent	16	6
medIncome	50670	13871
popEst2015	451131	1194160
incidenceRate	454	47
avgAnnCount	1990	4689
fiveYearTrend	-2	3
avgDeathsPerYear	731	1717

(Figure 8.1)

Method

Categorical predictor coding (1, 0)

Continuous Predictor Standardization

Subtract the mean, then divide by the standard deviation

Predictor	Mean	StDev
PovertyEst	70542.0	211675
incidenceRate	453.9	47
fiveYearTrend	-1.8	3
medIncome	50671.7	13871

(Figure 8.2)

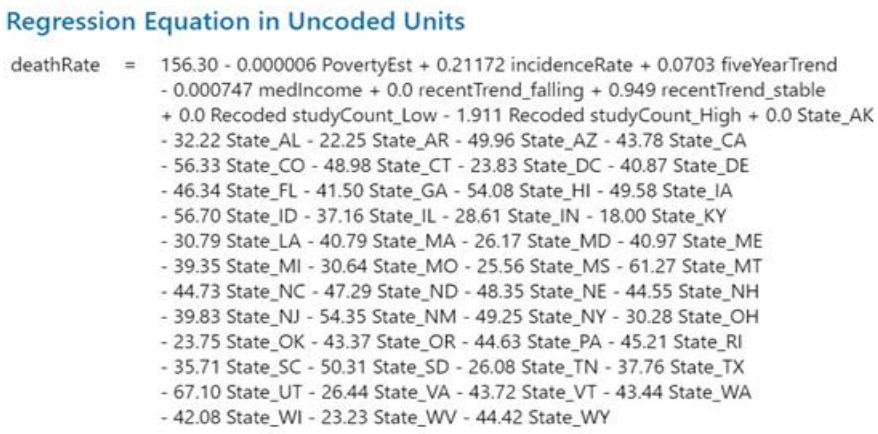
Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
14.9314	63.45%	63.39%	63.31%

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	53	11723701	221202	992.18	0.000
PovertyEst	1	37695	37695	169.08	0.000
incidenceRate	1	1832542	1832542	8219.69	0.000
fiveYearTrend	1	1210	1210	5.43	0.020
medIncome	1	1972855	1972855	8849.05	0.000
recentTrend	1	3203	3203	14.37	0.000
Recoded studyCount	1	7197	7197	32.28	0.000
State	47	2995144	63726	285.84	0.000
Error	30287	6752347	223		
Lack-of-Fit	3527	6752347	1914	*	*
Pure Error	26760	0	0		
Total	30340	18476048			

(Figure 8.3)



We used the identical techniques to create the "Death Rate" Regression model as we did for the Incidence Rate model. (Refer to the image above and Appendix, Table 3)

Based on these findings, we may conclude that there is no statistically significant association between the ***poverty estimate*** and the death rate. The slope coefficient of 0.000006 indicates that as the poverty estimate rises by 1%, the cancer death rate rises by 0.000006% while all other factors remain constant. There is a considerable positive link between the mortality rate and the ***incidence rate***. The slope coefficient of 0.21172 indicates that while the incident rate rises by 1%, the Death Incident rate rises by 0.217% while all other variables remain constant. The ***five-year trend*** is also significantly related to the mortality rate. Holding all variables equal, the mortality rate is predicted to rise by 0.0703% as the five-year trend rises by one year. The link between ***MedIncome*** and mortality rate is not statistically significant. The slope coefficient is -0.000747, indicating that with every unit increase in MedIncome, the cancer death rate is anticipated to reduce by 0.000747% when all other factors remain constant.

8. Conclusion and Recommendations

In summary, our study aimed to investigate the factors that influence cancer incidence rates in the United States. We conducted various analyses, including descriptive statistics, hypothesis tests, ANOVA, and correlation analysis, to explore the association between different variables and cancer incidence rates. Our findings revealed that states such as Kentucky, Delaware, and New York have the highest average number of cancer incidents, while Arizona, New Mexico, and Utah are at the bottom of the list. Furthermore, we found that recent trends and study counts had a significant impact on the incidence rate. In states with a falling trend and a high number of studies, the mean incidence rate was lower than in states with a stable trend and a limited number of studies.

We also investigated the role of median income, and our analysis revealed that the means of the four populations were not comparable. The differences between all these means were significant. Additionally, we identified independent factors that are strongly connected with cancer incidence rates and highly correlated with each other. By excluding these variables from the regression model, we were able to develop a more accurate model for the incidence rate and mortality rate separately.

In conclusion, our study provides valuable insights into the factors that influence cancer incidence rates in the United States. We recommend that future research investigates the unique geographical factors that may play a role in cancer growth. Furthermore, we encourage government organizations, medical institutions, and the general public to implement effective cancer prevention strategies based on our findings.

Appendix

Table 1

Statistics									
Variable	**State**	**N**	**N***	**Mean**	**SE Mean**	**StDev**	**Minimum**	**Median**	**Maximum**
incidenceRate	AK	178	0	421.25	5.24	69.96	201.3	441.4	501.1
	AL	639	0	459.33	1.04	26.32	391.8	460.7	555.6
	AR	588	0	435.6	1.53	37.01	364.1	433.2	532.1
	AZ	402	0	369.25	2.15	43.11	269.9	379.1	444.9
	CA	1750	0	427.89	0.621	25.98	312.8	420.3	490.8
	CO	512	0	399.14	1.93	43.66	234	408	454.7
	CT	279	0	482.5	1.1	18.35	431.4	488.8	500.1
	DC	31	0	483.7	0	0	483.7	483.7	483.7
	DE	67	0	498.19	2.04	16.68	485.9	493.3	527.2
	FL	980	0	436.95	1.88	58.8	259.6	430.8	1206.9
	GA	723	0	456.93	1.42	38.2	295.7	463	550.7
	HI	92	0	411.56	1.35	12.91	392.2	418.3	425.2
	IA	931	0	468.79	0.924	28.19	390.3	470.8	548.4
	ID	273	0	431.54	2.88	47.66	345.3	435.5	538.6
	IL	1380	0	482.91	0.719	26.7	392.3	480.1	551.5
	IN	770	0	453.15	0.937	26	396.8	451.1	535.7
	KS	695	0	453.55	0	0	453.55	453.55	453.55
	KY	760	0	517.33	1.25	34.56	343.2	518.8	639.7
	LA	512	0	486.81	1.25	28.27	379	487.1	586.2
	MA	534	0	477.32	1.27	29.32	413.9	473.1	572.8
	MD	466	0	452.39	1.78	38.47	392.9	457	532.1
	ME	432	0	480.25	0.957	19.89	435.2	479.6	507.3
	MI	974	0	448.98	1.8	56.25	310.1	463.9	541.3
	MN	880	0	453.55	0	0	453.55	453.55	453.55
	MO	1015	0	442.57	1.44	45.75	287.4	451.6	529.5

	MS	417	0	471.22	2.08	42.5	339	468.1	561.9
	MT	348	0	450.37	2.42	45.21	348.1	446.65	587
	NC	803	0	459.46	1.1	31.28	370.9	457.6	524.4
	ND	376	0	440.19	2.97	57.63	254.7	456.9	579.7
	NE	559	0	434.05	1.81	42.81	221.5	431.1	529
	NH	248	0	485.15	1.66	26.1	435.5	482.2	518.8
	NJ	594	0	494.01	1.41	34.34	399.5	493.1	559.1
	NM	364	0	375.17	2	38.12	303	366.5	558.5
	NV	174	0	453.55	0	0	453.55	453.55	453.55
	NY	1767	0	497.52	0.669	28.14	435.2	500.8	577.4
	OH	1191	0	452.34	0.906	31.27	316.5	458.7	514.7
	OK	644	0	442.32	1.42	35.99	354.7	449.4	516.3
	OR	415	0	442.77	1.6	32.61	358.5	441.7	587.4
	PA	1787	0	481.62	0.601	25.41	393.5	484	560.4
	RI	77	0	479.71	1.09	9.56	469	473.3	492.3
	SC	420	0	451.81	1.29	26.36	383	459	497.9
	SD	343	0	436.01	3	55.65	284.8	426.4	577
	TN	617	0	469.9	1.1	27.25	407.4	470.6	538.7
	TX	1902	0	411.91	0.944	41.15	211.1	424.25	537.2
	UT	278	0	398.96	2.19	36.5	261.1	404.1	489.9
	VA	871	0	427.37	2	58.98	265.7	421.6	1014.2
	VT	254	0	467.04	1.2	19.13	436.9	463.7	548.5
	WA	592	0	452.26	1.84	44.79	293.6	461.8	518.4
	WI	768	0	447.84	1.69	46.73	278.5	455	572.8
	WV	703	0	469.87	1.74	46.22	234	469.6	591
	WY	176	0	407.56	2.96	39.31	280.9	411.3	484.2

Table 2

Term	Coef	SE Coef	T-Value	T-Value	VIF
Constant	394.89	2.6	152.04	0	
studyCount	0.261	0.201	1.3	0.194	1.23
PovertyEst	21.94	1.91	11.49	0	111.05
povertyPercent	-6.452	0.386	-16.71	0	4.54
medIncome	2.604	0.528	4.93	0	8.48
popEst2015	-73.52	2.9	-25.31	0	256.8
avgAnnCount	208.22	6.12	34.03	0	1139.91
fiveYearTrend	3.652	0.199	18.31	0	1.21
deathRate	26.488	0.274	96.77	0	2.28
avgDeathsPerYear	-155.18	5.41	-28.68	0	891.36
State					
AL	53.35	2.77	19.27	0	4.81
AR	24.39	2.8	8.71	0	4.54
AZ	12.79	3.04	4.21	0	3.67
CA	46.22	2.63	17.6	0	11.41
CO	31.6	2.9	10.9	0	4.15
CT	73.1	3.14	23.27	0	2.74
DC	69.48	6.19	11.22	0	1.19
DE	83.02	4.58	18.12	0	1.41
FL	43.75	2.72	16.08	0	7.05
GA	60.45	2.76	21.93	0	5.02
HI	33.94	4.15	8.19	0	1.58
IA	65.74	2.73	24.12	0	6.28
ID	50.55	3.18	15.9	0	2.71
IL	67.48	2.62	25.76	0	9.02
IN	38.74	2.72	14.25	0	5.57
KY	81.71	2.71	30.14	0	5.46
LA	72.45	2.84	25.54	0	3.91
MA	68.98	2.84	24.32	0	4.24
MD	46.53	2.87	16.22	0	3.79
ME	71.83	2.92	24.6	0	3.64
MI	42.82	2.69	15.91	0	6.75

MO	35.6	2.68	13.3	0	6.96
MS	59.38	2.92	20.33	0	3.49
MT	71.48	3.1	23.05	0	2.99
NC	63.24	2.72	23.22	0	5.82
ND	50.97	3.03	16.83	0	3.11
NE	40.83	2.86	14.27	0	4.39
NH	79.68	3.2	24.91	0	2.52
NJ	77.46	2.81	27.54	0	4.42
NM	10.31	3.07	3.36	0.001	3.25
NY	81.33	2.61	31.15	0	11.38
OH	41.85	2.64	15.88	0	7.97
OK	29.88	2.76	10.83	0	4.82
OR	51.6	2.93	17.61	0	3.5
PA	72.77	2.6	27.93	0	11.45
RI	77.96	4.37	17.85	0	1.47
SC	48.31	2.93	16.51	0	3.47
SD	48.68	3.09	15.74	0	2.92
TN	52.86	2.77	19.07	0	4.66
TX	27.19	2.59	10.5	0	11.7
UT	33.37	3.24	10.3	0	2.7
VA	26.9	2.69	10.01	0	6.07
VT	62.44	3.18	19.63	0	2.56
WA	51.21	2.8	18.27	0	4.57
WI	45.55	2.74	16.61	0	5.65
WV	50.26	2.76	18.24	0	5.15
WY	19.47	3.46	5.63	0	2.1
recentTrend stable	-0.494	0.543	-0.91	0.363	1.59
Recoded studyCount High	6.412	0.785	8.17	0	1.28
Recoded medIncome					
1	8.768	0.647	13.54	0	2.4
2	12.62	0.821	15.38	0	3.84
3	7.88	1.14	6.91	0	7.41

Table 3

Term	Coef	SE Coef	T - Value	P - Value	VIF
Constant	215.12	1.16	186.18	0	
studyCount	0.1283	0.0903	1.42	0.155	1.23
PovertyEst	-21.319	0.851	-25.04	0	109.29
povertyPercent	3.78	0.173	21.85	0	4.51
medIncome	-5.61	0.235	-23.88	0	8.32
popEst2015	32.07	1.31	24.55	0	257.31
incidenceRate	10.339	0.108	95.44	0	1.77
avgAnnCount	-113.05	2.73	-41.47	0	1120.13
fiveYearTrend	0.2659	0.0901	2.95	0.003	1.22
avgDeathsPerYear	101.01	2.39	42.17	0	864.71
State					
AL	-32.3	1.24	-26.09	0	4.77
AR	-21.76	1.25	-17.35	0	4.51
AZ	-54.31	1.33	-40.83	0	3.49
CA	-43.85	1.16	-37.82	0	11.02
CO	-52.66	1.27	-41.46	0	3.95
CT	-43.14	1.4	-30.75	0	2.7
DC	-29.74	2.78	-10.69	0	1.19
DE	-37.44	2.06	-18.19	0	1.41
FL	-43.92	1.2	-36.52	0	6.8
GA	-40.09	1.23	-32.66	0	4.94
HI	-50.63	1.84	-27.48	0	1.55
IA	-42.96	1.21	-35.45	0	6.16
ID	-51.09	1.4	-36.37	0	2.62
IL	-34.19	1.17	-29.13	0	8.98
IN	-24.53	1.22	-20.14	0	5.53
KY	-18.55	1.23	-15.05	0	5.59
LA	-30.84	1.28	-24.16	0	3.92
MA	-39.02	1.27	-30.78	0	4.19
MD	-28.54	1.28	-22.22	0	3.76
ME	-36.8	1.31	-28.12	0	3.63
MI	-33.93	1.2	-28.3	0	6.63
MO	-28.55	1.2	-23.89	0	6.87
MS	-29.32	1.32	-22.28	0	3.46
MT	-56.44	1.37	-41.25	0	2.88

NC	-41.84	1.21	-34.54	0	5.7
ND	-41.54	1.35	-30.85	0	3.05
NE	-41.3	1.27	-32.58	0	4.27
NH	-40.31	1.43	-28.12	0	2.51
NJ	-37.66	1.26	-29.85	0	4.41
NM	-53.3	1.35	-39.59	0	3.1
NY	-39.97	1.17	-34.15	0	11.33
OH	-27.87	1.18	-23.65	0	7.9
OK	-21.91	1.24	-17.72	0	4.79
OR	-41.37	1.3	-31.76	0	3.43
PA	-40.8	1.16	-35.09	0	11.3
RI	-43.44	1.96	-22.2	0	1.46
SC	-34.18	1.31	-26.17	0	3.42
SD	-44.51	1.37	-32.44	0	2.84
TN	-25.18	1.24	-20.23	0	4.65
TX	-35.87	1.15	-31.24	0	11.38
UT	-60.38	1.42	-42.61	0	2.56
VA	-24.94	1.2	-20.77	0	6.01
VT	-37.25	1.42	-26.18	0	2.53
WA	-38.86	1.25	-31.17	0	4.48
WI	-36.2	1.22	-29.65	0	5.54
WV	-22.74	1.24	-18.36	0	5.16
WY	-38.87	1.54	-25.28	0	2.06
recentTrend					
stable	1.598	0.244	6.55	0	1.59
Recoded studyCount					
High	-2.508	0.353	-7.11	0	1.28
Recoded medIncome					
1	-5.447	0.29	-18.78	0	2.38
2	-7.405	0.368	-20.14	0	3.82
3	-5.229	0.512	-10.22	0	7.39